

TECHNICAL BULLETIN



MODEL/DERIVATIVE:

All Models

Bull. N°: **0013**

CDS. ref **R6499bu**

Issue: **1**

Date: **21.08.96**

AFFECTED RANGE:

All fuel injected petrol engine vehicles fitted with closed loop catalyst.

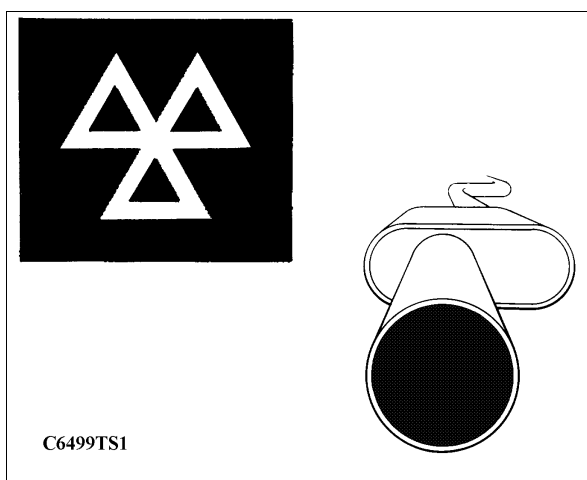
PROBLEM:

EXHAUST EMISSIONS FAILURE, VARIOUS - due to any of following:-

- High Lambda reading.
- High CO reading at idle / fast idle.
- High HC reading.
- Incorrect idle/fast idle.

This bulletin is primarily intended as a diagnostic aid for vehicles which have failed the UK ministry of Transport emissions test, but it may also help diagnose engine running faults.

All emission readings provided are the test limits currently agreed for UK vehicles with standard 95 Ron fuel.



CAUSE:

Any of the following:-

- Exhaust leak.
- Rich mixture.
- Rich mixture with misfire.
- Weak mixture.
- Oxygen sensor inoperative.
- Catalyst fault.
- Miscellaneous MEMS system faults.

ACTION:

INTRODUCTION:

This bulletin should be used with reference to workshop poster D6499TS1, a copy of which is included with this bulletin. A reduced size copy is also included for bulletin record purposes.

Following emissions test failure, the car should be re-checked in accordance with the top row of boxes on the poster and the readings recorded. The boxes include Rovers recommended readings. Having established which readings are outside recommended values, the 'Result' boxes on the lower part of the poster should be applied along with rectification advice suggested later in this bulletin, see 'Result Action' section.

CHECKS AND READINGS EXPLAINED (POSTER):

From left to right - upper row

1. Oil temperature is not only a requirement of the UK ministry test but is considered necessary to confirm engine is fully warm and provide accurate engine readings, coolant temperature alone is not sufficient since coolant reaches its correct temperature prematurely.
 2. By raising engine speed to fast idle the fuelling changes from open loop to 'closed loop' (catalyst lit). Fast idle speed parameters vary dependent on MEMS tune, i.e. K 1.4 requires a lower parameter to prevent system inadvertently returning to an 'open loop' condition.
 3. CO (Carbon Monoxide) at fast idle is only relevant if found to be higher than limits. Lower readings can be ignored.
 4. HC (Hydrocarbons) readings are necessary to give indications as to a possible engine malfunction.
 5. Lambda is a calculation of the actual fuel ratio divided by a standard air fuel ratio, for example 16.1 (recorded AFR) divided by (14.7 (standard AFR) = 1.09 Lambda. Lambda may be considered as a variation around the ideal mixture value.
 6. Re-checking CO (Carbon Monoxide) at idle will provide a reading in an open loop condition, if CO is within limits this confirms catalyst is still efficient in this mode.
 7. O2 (Oxygen) reading obtained at idle. Reading provides a measure of oxygen content, relevant for diagnosis of certain conditions.
 8. Oxygen sensor voltage swing test to confirm correct oxygen sensor function.
- Voltage stuck high or low may indicate fuel mixture errors, see 'Poster Results' for further information.

RESULTS ACTION (POSTER)

EXHAUST LEAK:

Inspect complete exhaust system, including manifold for leaks. To exaggerate leak, partially block off tail pipe during inspection. Exhaust system joints may require re-sealing even if they appear intact, even minute leaks, i.e.

around clamped connection joints will have an adverse affect on emissions reading.

Note: If CO reading was high, this may indicate leak is before the oxygen sensor, i.e. exhaust manifold joint.

RICH MIXTURE

Possible causes:

- pressure high (manifold pressure).
- Valve timing incorrect.
- Injectors over-fuelling.
- High fuel pressure - kinked return pipe.

Run TestBook 'Healthcheck' to test complete MEMS system, view results and correct any failures.

RICH MIXTURE WITH MISFIRE

Possible causes:

- Valves burnt.
- Valves bent or seating badly (see Note:).

Check with KV tester or oscilloscope to diagnose HT related causes. If no HT faults are found, check cylinder compression or carry out cylinder leakage test. Rectify electrical or mechanical cause as necessary.

Note: Condition may be due to valve sticking, refer to appropriate Technical Bulletins for advice. 'K' Series - Bulletin 0001, (engine section) or 'T' Series - Bulletin 0015 (engine section).

WEAK MIXTURE

Possible causes:

- Fuel pressure too low (e.g. Blocked fuel filter)
- Blocked fuel injector.

Run TestBook 'Healthcheck' to test complete MEMS system, view results and correct any failures.

OXYGEN SENSOR INOPERATIVE

Oxygen sensor testing;

The oxygen sensor operation can be tested via TestBook. If unavailable, use Tool 18G1639. Either can be used to view an actual voltage swing to confirm correct oxygen sensor function.

Note: An added advantage with TestBook is the ability to view other related information, on the same screen, i.e., engine speed, throttle angle, MAP reading, stepper steps, ignition advance, coolant temp, inlet air temp, battery volts, ignition switch and injector dead time.

Oxygen Sensor Check - TestBook:

From the welcome screen , select - 'Diagnostic System'.

When requested select appropriate MEMS system.

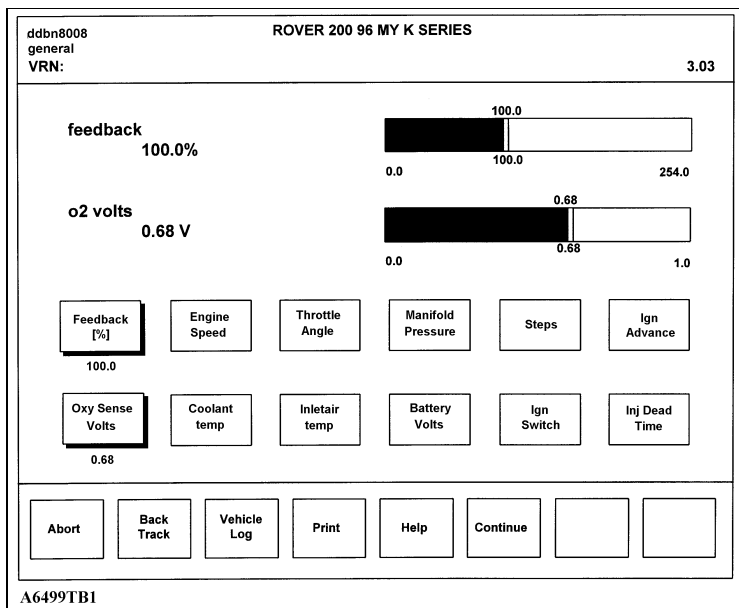
Proceed as instructed until menu screen appears, select - 'Sub System Tests'.
From this screen select - 'Display Outputs'.
From this screen select - 'General'.
Select OxySense Volts and Feedback % (see illustration 2).

OxySense Volts - See 'Test Results' later in this bulletin for acceptable readings.

Feedback - Display of feedback fuelling correction. This is shown as a percentage of the mapped (open loop) value. Percentage is continuously updated by the MEMS ECM whenever the conditions for closed loop fuelling are present.

High values of feedback (e.g. above 100%) indicate that feedback is attempting to compensate for fuelling being too lean. Lean values (e.g. below 100%) indicate fuelling being too rich.

Note: Misfire condition will be shown as high values as feedback will be fooled into compensating for a system running too lean.



Oxygen Sensor Check - Tool 18G1639 (alternative method)

A voltmeter may be connected between the sensor and engine harness connection.

A tee-in tool is available for this purpose, part number:-
18G 1639 (for square type engine harness connections)

Procedure:

Ensure engine is at correct running temperature.

Switch off engine and connect 'T' lead 18G 1639 between oxygen sensor and engine harness.

Connect 'T' lead voltmeter connections, yellow to voltmeter negative and green to voltmeter positive.

Oxygen Sensor Check - Method (for both TestBook and Tee-in tool use):

Increase engine speed to 3000 r.p.m for approximately 30 seconds then allow to idle, this is necessary to ensure oxygen sensor is at operating temperature. If voltage swings regularly between below 0.4 and above 0.6 volts, sensor is operating correctly. If reading is stuck high or low (see associated readings detailed on poster) this will indicate sensor is not working.

Oxygen Sensor voltage stuck high or low:

If engine has been idling or running at low rpm for long periods prior to test, sensor may have become sooted, road test with engine speed held high in an attempt to clean sensor without removal.

If above fails to resolve condition, remove sensor and view tip, if still sooted, carefully clean and refit to vehicle for re-test.

If sensor voltage refuses to swing and remains high, fault may be rich mixture. Check fuel return lines for kinks or restrictions or carry out fuel pressure check.

If fuel system faults are eliminated and sensor refuses to swing correctly, replace sensor.

Oxygen sensor testing - General:

If the system is working properly, the meter reading will switch regularly between below 0.4v and above 0.6v. The actual meter readings are not too important, the fact that it oscillates between low and high values is enough to prove the system is working and hence, that fuelling is unlikely to be the cause of any running problems.

If during the test, readings did not oscillate but remained high, the mixture is over rich. Check for kinks or restrictions in the fuel return line or use TestBook to diagnose the fuel system.

A constantly low reading could be due to one of two things, either the fuelling is too weak, or the oxygen sensor is not working. With the engine running, clamp the fuel return line for a maximum of 10 seconds, this will create an over rich mixture which in turn should prompt a high voltage reading (above 0.6v) from the oxygen sensor. If this is the case then check for a blocked fuel filter, kinked fuel feed line or any other possible causes.

Note: Fuel hoses must be clamped for a short periods with approved Girling brake hose clamp No. 64947017.

CATALYST FAULT

It is unlikely that the catalyst has failed unless it has been subjected to an unchecked misfire condition or incorrect fuel use. Since a new catalyst is a very expensive component it should only be replaced as a last resort.

A catalyst which has completely failed will on removal from the vehicle either deposit loose matrix particles and when shaken be heard to rattle in casing or if viewed through open end show that matrix is blocked or melted.

MISCELLANEOUS RESULTS

Run TestBook 'Healthcheck' to test complete MEMS system, view results and correct any failures. Healthcheck can be accessed via the following route:

From the welcome screen , select - 'Diagnostic System'.

When requested select appropriate MEMS system.

Proceed as instructed until menu screen appears, select - 'Healthcheck'.

CONCLUSION

Applying poster:

- Identify emission irregularities by comparing actual readings with those from poster.
- Compare result combinations with those from poster for likely cause.

Recommended diagnostic tools:

- Use TestBook (display outputs) or Tool 18G1639 for checking voltage swings.
- Use TestBook (Healthcheck) for general MEMS check.
- Use oscilloscope to diagnose HT faults.
- Use compression tester or leakage tester for checking engine condition.

PARTS INFORMATION:

Replace as required - not warranty

WARRANTY CLAIMS:

Not applicable.

